

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

AU-701-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mechanical Vibration

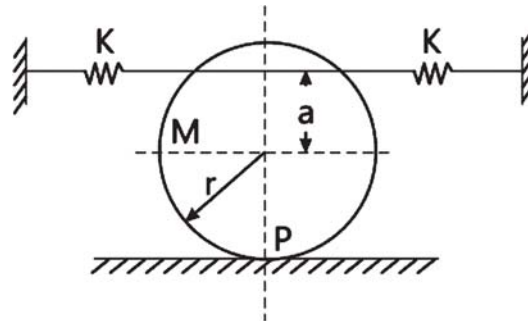
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Determine the natural frequency of the system shown in the Fig. by Newton's method and energy method. 7



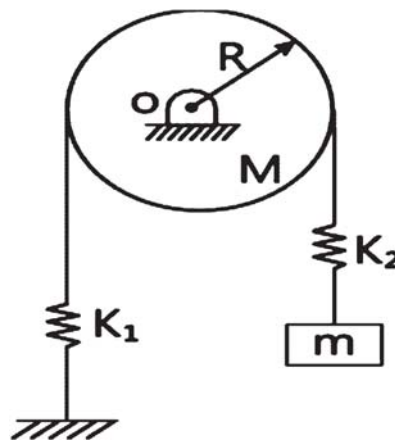
- b) Set up the differential equation for a spring mass damper system and obtain complete solution for the critically damped condition. 7
2. a) Define transmissibility and derive an expression for the transmissibility ratio and the phase angle for transmitted force. 7

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- b) Determine the natural frequencies of the spring mass pulley system as shown in the Fig. The cord is inextensible and there is no slippage between the cord and the pulley. Take $K_1=40\text{N/m}$, $K_2=60\text{N/m}$; $m=2\text{kg}$ and $M=10\text{kg}$. 7



3. a) A 5 kg mass attached to the lower end of a spring, where upper end fixed, vibrates with a natural period of 0.45 sec. Determine the natural period when a 2.5 kg mass is attached to the midpoint of the same spring with upper and lower ends fixed. 7
- b) A shaft supported freely at the ends has a mass of 100 kg placed 25 cm from one end. Find the frequency of the natural transverse vibration if the length of the shaft is 75 cm, $E = 200 \text{ GN/M}^2$ and shaft diameter is 4 cm 7
4. a) What do you understand by under - damped system, over damped system and critically - damped system and its use? Explain. 7
- b) A vibratory system in a vehicle is to be designed with the following parameters: $K = 100\text{N/m}$, $c = 2\text{N- Sec/m}$, $m = 1 \text{ kg}$ calculate the decrease of amplitude from its starting value oscillations and the frequency of oscillation. 7

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5. a) A spring mass damper system is subjected to a harmonic force. The amplitude is found to be 20 mm at resonance and 10 mm at a frequency 0.74 times the resonant frequency. Find the damping ratio of system. 7
b) Explain the principle of undamped dynamic vibration absorber. 7
6. a) Write short notes on Vibration Isolation. 7
b) Explain Torsional vibration in detail. 7
7. a) Write short notes on vibration measurement. 7
b) Find the natural frequency of three mass spring system connected in series and hanging vertically. 7
8. a) What do you mean by Two degree of freedom vibration system. 7
b) What assumption are made in finding the natural frequency of single degree of freedom using energy method. 7
